

Truck Packing Program

1 Problem Description

A logistics company has asked us to provide them a program to support them in finding the best way for them to pack their trucks to maximise their profits. They have a number of trucks which need to deliver to multiple distribution centers around the UK area. The company regularly gets new vehicles and the stock is constantly changing. The company has given us a sample of the spreadsheets which they use to plan the truck loads manually.

The company would like a program that will maximise the amount of profit they get when they send out a truck to ensure shareholder value for money. The company would like the program to work on one truck at a time as they come into the depot. The depot will have more items than can fit in one truck.

The company has mentioned that they only deliver one size of box, but each box has a value and a weight and that these are not always linked.

2 Deliverables

Individual Truck Loading Program They would like a visual program to load up a chosen truck based on their current inventory.

Documentation They would also like documentation on how to use the program provided. They would like this as a PDF document.

3 Example Data

Table 1: Example Truck Data

Truck Type	Example Truck	Max Load Weight
Small Truck	Suzuki Supercarry	740KG
Medium Truck	Ford Transit	1500KG
Large Truck	MAN TGX	22000KG

Table 2: Example Item Data

ItemID	Weight	Profit
I1	60KG	£50
I2	150KG	£125
I3	2300KG	£1002
I4	125KG	£250